

- 14** The painkiller ibuprofen reduces the activity of an enzyme involved in the production of prostaglandin.

Ibuprofen binds reversibly to the active site of the enzyme.

Which type of enzyme inhibition describes this example and why?

- A** competitive inhibition because ibuprofen alters the shape of the enzyme
- B** competitive inhibition because ibuprofen and the substrate cannot bind at the same time
- C** non-competitive inhibition because ibuprofen blocks the active site
- D** non-competitive inhibition because ibuprofen has the same shape as the substrate